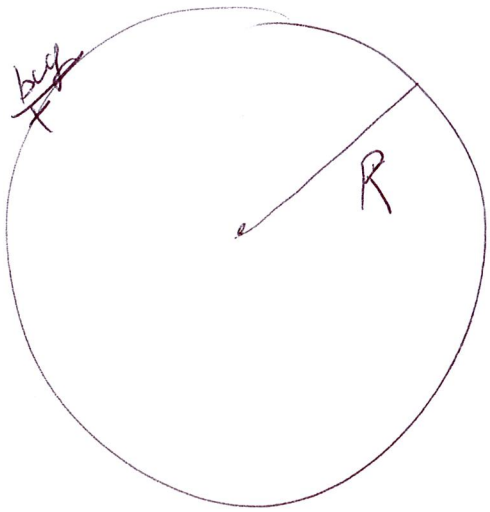
- A turntable of radius R spins at a constant rate, completing a spin in time T.

Find the:

1. - magnitude of angular velocity
2. - magnitude of angular acceleration
3. - linear speed of a bug on the edge of the turntable.

Rotational Motion Symbolic practice



The door below has length L

and mass M.

The rotational inertia is $\frac{1}{3}ML^2$

The bug has negligible mass.

Find

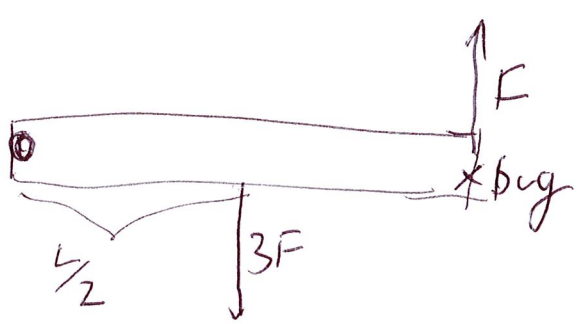
4. The net torque (magnitude + direction)

5. The ~~ang~~ angular acceleration (magnitude + direction

6. Angular speed after time Δt $(\omega_0 = 0)$

7. Liner speed of a bug on the edge after time Δt

8. - Angular momentum (magnitude + direction) after time ~~to~~ Δt.



9. Kinetic energy after time Δt.

A planet of mass M ~~sp~~ and radius R has a length of day T.

Rotational inertia of a sphere is $\frac{2}{5} M R^2$

Find

(10) - Angular speed

(11) - Angular Momentum

(12) - Angular kinetic energy

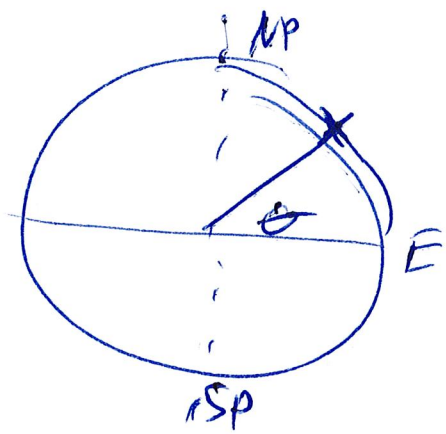
(13) - Linear speed of a person at the equator.

(14)* - Linear speed of a person at latitude θ.

(- use trigonometry to calculate moment arm.

$V = r\omega$)

Latitude



E - equator

NP - north pole

SP - south pole

the ~~lat~~ latitude of the person standing at point X is θ.

A hoop wheel of radius R and mass M moves at <u>linear</u> speed V.

Its rotational inertia is MR^2.

~~(15)~~ Calculate:

(15) Its angular speed

(16) Its ~~ang~~ angular momentum

(17) Its rotational kinetic energy

(18) Its <u>total</u> kinetic energy
(translational KE + rotational KE)

(19 bonus) Reason out why the rotational inertia is MR^2.
Would that be the rotational inertia of a uniform disk of the same mass and radius?

Answers

1. $\omega = \frac{2\pi}{T}$

2. $\alpha = 0$

3. $v = R\omega = \frac{2\pi R}{T}$ Oh look! $\frac{\text{circumference}}{\text{time}} = \frac{\text{distance}}{\text{time}}$

4. $\frac{LF}{2} \otimes = \cancel{\otimes} \Sigma\tau$

5. $\alpha = \frac{\Sigma\tau}{I} = \left(\frac{LF}{2}\right)^{\otimes} \frac{1}{\frac{1}{3}ML^2} = \frac{3}{2}\frac{F}{ML} \otimes \quad \frac{kg \cdot m/s^2}{kg \cdot m} \rightarrow \frac{1}{s^2}$ ✓

6. $\omega = \omega_0 + \Delta t\alpha = 0 + \Delta t\alpha = \frac{3F\Delta t}{2ML}$

7. $v = \frac{3}{2}\frac{F \cdot \Delta t}{M}$

Answers

(8) $L = I\omega = \frac{1}{3} ML^2 \times \frac{3F\Delta t}{2ML} = \frac{1}{2} FL\Delta t$

(9) $K_r = \frac{1}{2} I\omega^2 = \frac{1}{2} \times \frac{1}{3} ML^2 \times \left(\frac{3}{2}\frac{F\Delta t}{ML}\right)^2$

$= \frac{3}{8} \cdot \frac{\cancel{ML^2}\, F^2 \Delta t^2}{M^2 \cancel{L^2}} = \frac{3}{8} \frac{F^2 \cdot \Delta t^2}{M}$ $\quad \frac{kg^2 \cdot \frac{m^2}{s^4} \cdot s^2}{kg} \rightarrow kg\frac{m^2}{s^2} = Joule$ ✓

(10) $\omega = \frac{2\pi}{T}$

(11) $L = I\omega = \frac{2}{5} \cdot MR^2 \cdot \frac{2\pi}{T} = \frac{4\pi}{5} \cdot \frac{MR^2}{T}$

(12) $K_r = \frac{1}{2} I\omega^2 = \frac{1}{2} \cdot \frac{2}{5} \cdot MR^2 \cdot \left(\frac{2\pi}{T}\right)^2 = \frac{4\pi^2}{5} \cdot \frac{MR^2}{T^2}$ $\quad kg \cdot \frac{m^2}{s^2}$ ✓

Answers

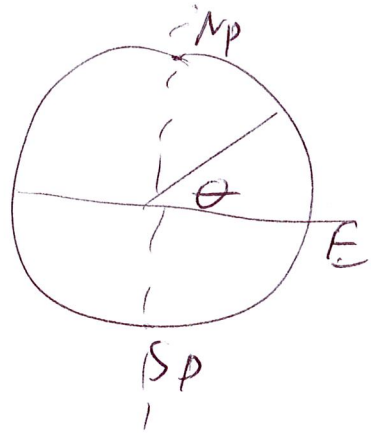
(13)

$$V = R\omega = \frac{2\pi R}{T}$$

(14)

$$\cancel{V = \frac{2\pi R \sin(\theta)}{T}}$$

$$V = \frac{2\pi R \cos(\theta)}{T}$$



latitude = θ

(15) $\omega = \frac{V}{R}$

(16) $L = I\omega = MR^2 \cdot \frac{V}{R} = MVR$

(17) $K_r = \frac{1}{2} I\omega^2 = \frac{1}{2} MR^2 \left(\frac{V}{R}\right)^2 = \frac{1}{2} MV^2$

(18) $K_{tot} = K_{trans} + K_{rot} = \frac{1}{2} mv^2 + \frac{1}{2} mv^2 = mv^2$

$K_{tot} = mv^2$